TITLE : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Conclusion

This Java program successfully demonstrates the implementation of the final keyword across variables, methods, and classes to enforce immutability and restrict inheritance. By declaring critical attributes like account numbers and ISBNs as final, the application guarantees that sensitive identifiers cannot be altered after initialization. Furthermore, applying final to methods prevents unintended method overriding in subclasses, while declaring a class as final restricts inheritance entirely to safeguard structural design. Overall, the final keyword enhances security, avoids accidental state modification, and improves program reliability.

Exercise

1. Create a **Bank Account program** where account number is final and cannot be changed.

Code:

```java
class BankAccount {
    private final String accountNumber;
    private String holderName;
    private double balance;
    public BankAccount(String accountNumber, String holderName, double balance) {
        this.accountNumber = accountNumber;
        this.holderName = holderName;
        this.balance = balance;
    }
    public void displayAccountInfo() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Holder Name: " + holderName);
        System.out.println("Balance: $" + balance);
    }
    public static void main(String[] args) {
        BankAccount account = new BankAccount("ACC-987654", "John Doe", 2500.0);
        account.displayAccountInfo();
    }
}
```

Output:

```
Listening on 61102
User program running
Account Number: ACC-987654
Holder Name: John Doe
Balance: $2500.0
User program finished
```

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Code:

```java
final class Book {
    private final String isbn;
    private String title;
    private String author;
    private double price;
    public Book(String isbn, String title, String author, double price) {
        this.isbn = isbn;
        this.title = title;
        this.author = author;
        this.price = price;
    }
    public final void displayBookDetails() {
        System.out.println("=== Book Details ===");
        System.out.println("ISBN: " + isbn);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: $" + price);
    }
    Run main | Debug main
    public static void main(String[] args) {
        Book book = new Book("978-0-13-468599-1", "Effective Java", "Joshua Bloch", 45.00);
        book.displayBookDetails();
    }
}
```

Output:

```
Listening on 54438
User program running
=== Book Details ===
ISBN: 978-0-13-468599-1
Title: Effective Java
Author: Joshua Bloch
Price: $45.0
User program finished
```